Vehicle Log Book entry procedure

Review of VLB Manual, VLB Scan and PDF upload, with suggested changes

Caliber Pulse MIS | Repository main at commit d7486f8 | Prepared 2 November 2026 | For team review, no changes made to the code

1. Summary

All three ways of entering a Vehicle Log Book (VLB) card end in the same server path: the card must belong to an open batch for its site, date and shift, the batch must be in the VLB stage, the date must be on or after 1 November 2026, and the card number is allocated inside the save transaction. That core is sound. Card numbering has no gaps or races, permission checks are repeated on the server, and the upload endpoints validate inputs well.

The weaknesses are in what happens before the save. The entry screens let a user into a batch that will refuse every save, the scan setup never confirms that a VLB batch exists until the last step, and completed scan cards are posted automatically although the written procedure says every card is approved by a person. Each of these costs the MIS user time, and the third one needs a team decision.

Headline recommendations

- Filter the batch picker on stage and amendment owner, not only on closed status (finding 1).
- Check for an open VLB batch when a scan batch is created, before pages are uploaded and read (finding 2).
- Decide whether auto-save of complete scan cards is wanted; then align either the code or the documentation (finding 3).
- Fix the shift lookup so the Manual form cannot open with an empty, disabled Shift field (finding 4).

2. Procedure as implemented

2.1 Common rules for every VLB card

- An active batch must exist for the same site, report date and shift. Batches are created on the Site batches page by an MIS user or administrator. One open batch per site and one open batch per user.
- The batch must be in stage "VLB open". Once VLB is locked (stage "ELB open") or the batch is closed, no VLB card can be added or edited.
- If a batch was reopened after PM approval, only the approved MIS user may write to it.
- Report dates before 1 November 2026 are rejected by every path.
- All of this is enforced in assertVlbBatchWrite (vlb-batch-policy.mjs) inside the save transaction, together with the card number allocation (VLB_M for manual, VLB_S for scan, counted per date, shift and source).

2.2 VLB Manual

1. Vehicle Log Book page offers VLB Manual, VLB Scan and Site batches.
2. VLB Manual opens the batch picker. Choose a site; if exactly one open batch exists it opens automatically. If none exists the user is told to create it on Site batches.
3. The form opens with location, book date, shift and shift hours fixed from the batch and read-only.
4. The user enters door, work order, employee, A-Form number, opening and closing KMR and HMR, trip rows (loading machine, trips, time), optional job, breakdown, attributes and attachments.
5. Save checks that the entry still matches the batch context and that trip equipment is at the site, then posts to the log book API.
6. The server validates masters, employee at site, asset kind, shift and work order site, applies breakdown hours, checks the batch, allocates VLB_M/DDth-MMMyy/Shift/NNN, inserts the card and snapshots it into the batch with an audit entry.

7. Editing a saved card reruns the same checks; site, date and shift of a batched card cannot change.

2.3 VLB Scan and PDF upload

1. VLB Scan opens the same batch picker, then the scan page with Scan setup locked to the batch date, site and shift.

2. Pages come from the Windows document scanner (local connector), the cloud scanner connector, or PC files (PDF or image). Camera capture is not part of the scan flow.

3. PDFs are rendered in the browser with pdfjs: at most 120 pages per file, scaled to 2000 by 2667 pixels, JPEG, up to two pages per card, three uploads in flight at a time. A new scan entry is started every 40 cards. The local scanner path renders on the server instead, at 2400 by 3200 PNG, with 1 to 20 files and 12 MB per upload.

4. Each upload is hashed; exact duplicates of an unrejected or posted card are skipped.

5. Extraction runs one card at a time (Gemini hybrid when enabled, else PaddleOCR evidence, else local OCR).

6. A context gate holds the batch while any card is still reading or any card disagrees with the batch date, site or shift. The user must proceed, discard the wrong cards, or cancel.

7. The reviewer corrects fields, ticks the checked box and approves. Drafts autosave every 1.2 seconds.

8. Cards reach the log book three ways: Save all (every card approved and checked), the Save button on one card, or automatic saving of any card whose checks are clean. All three call the same insert path, which checks the batch and allocates VLB_S/DDth-MMMyy/Shift/NNN.

3. Findings and suggested changes

Severity reflects the cost to the MIS user or the risk to data, not the size of the fix. File references are to the repository at commit d7486f8.

1. Batch picker admits batches that cannot accept saves — HIGH

Where: `src/vlb-create-batch.jsx:152 (openVlbEntryBatches), src/vlb-entry-workspace.jsx:12`

Observed: The entry picker filters only on closed status. A batch already in the ELB stage, or one reopened for a different MIS user, opens the Manual form or the Scan page. The user completes a card, or uploads and reviews pages, and only then receives "VLB is locked for batch" or "reopened for another MIS user". The governance panel already hides Manual entry when the stage is not VLB open; the picker does not.

Suggested change: Filter on stage equals vlb_open and on amendment owner, and show the reason in the picker instead of an Open button. Also refresh the selected batch from the ten-second poll so a lock or closure by another user closes the form.

2. Scan batch creation never checks for an open VLB batch — HIGH

Where: `bulk-card-scanning.mjs:178 (POST /api/scan-batches), local-document-scanning.mjs:17`

Observed: Creating a scan batch validates the context format and site scope only. Through the cloud connector, the local scanner, or after a batch is locked mid-session, a full PDF can be uploaded, extracted and reviewed before the save reports that no batch exists or that VLB is locked. The VLB batch is checked only inside insertLogBook.

Suggested change: Call matchingVlbBatch with the stage check when a vehicle scan batch is created, and include the result in the context gate so the scan page shows "No open VLB batch for this site, date and shift" before upload.

3. Auto-save posts scan cards without approval, contrary to the documentation — HIGH

Where: `scan-auto-save.mjs:8 (canAutoSaveScanCard), src/bulk-card-scanning.jsx:165, SCAN-REVIEW-GROUPS.md:17 and :21`

Observed: A card in review status whose checks have no errors or blocking warnings is posted to the log book automatically, with the user shown "Saving complete card N automatically". The documentation states that no card is approved or posted automatically and that every card must be explicitly approved. The commit history shows auto-save was introduced deliberately, so the code and the written procedure disagree.

Suggested change: Team decision. Either require approved status in canAutoSaveScanCard, or keep auto-save, make it a visible per-batch setting, and correct SCAN-REVIEW-GROUPS.md and GEMINI-SCANNING.md.

4. Manual form can open with an empty, disabled Shift field — HIGH

Where: `src/vlb-entry-context.mjs:6 (vlbEntryFields), src/log-book-form.jsx:161, log-book.mjs:58`

Observed: The shift is matched against the Shift Master by code name only. When no row matches for that site and date, shiftId is empty, the Shift select is disabled so browser validation skips it, and the save fails on the server with "Shift is required". Pulse hours lookups are also silently disabled.

Suggested change: Reuse selectedMasterShift from trip-card-rules.mjs, which also matches on start time and handles dated revisions, and show "No Shift Master row for shift X at site Y" before the form opens.

5. Meter rule breaches are stored but not shown before save — MEDIUM

Where: `log-book.mjs:76, trip-card-rules.mjs:3, src/log-book-form.jsx:31`

Observed: Closing below opening, HMR above shift hours and KMR above the limit only populate meterRuleErrors on the saved card. The form shows a dash for a negative total and allows the save. Only Saved VLB surfaces the breach later. No duplicate-card check on door, date and shift exists in the manual path.

Suggested change: Show the breach inline next to the meter fields before save. Decide whether a negative total should block the save or only warn.

6. Manual entry cannot raise the awaiting-transfer hold — MEDIUM

Where: `src/log-book-form.jsx:97 and :178, log-book.mjs:90 and :106, docs/vlb-elb-batch.md section 6`

Observed: With two-stage batches on, the server accepts a loading machine from another site as a site hold. The Manual form still validates trips site-only and filters the equipment dropdown to the site, so a manual user can never trigger the hold and express transfer flow that the specification describes for manual cards.

Suggested change: Return the two-stage flag from the options endpoint, pass anySite to validateTripEquipment, and list other-site equipment in the dropdown with a transfer badge.

7. Edit, Delete and record navigation ignore batch state — MEDIUM

Where: `src/log-book-form.jsx:123, :125, :134, :145; log-book.mjs:238`

Observed: Edit is enabled for every saved card and fails only at save when the batch is locked or closed. The Delete button and dialog always end in a 403 because direct deletion is disabled. First and Last navigate the global register rather than the batch, so a card from another batch opens and the save is blocked with a generic message.

Suggested change: Disable Edit when the batch stage is not VLB open, remove Delete or route it to the Deletion page, and scope First and Last to the batch records.

8. Browser and server PDF paths use different limits, formats and messages — MEDIUM

Where: `src/bulk-card-scanning.jsx:54, src/scan-upload-image.mjs:3, local-document-scanning.mjs:11, :28, :31, :32`

Observed: Browser rendering uses 2000 by 2667 pixels at scale up to 2.5 as JPEG, 100 MB per source file and 12 MB per rendered card. The local scanner path renders 2400 by 3200 at scale up to 4 as PNG, with 12 MB for all files together. Page-limit messages differ. The local file upload path also lacks the whole-file hash dedupe that the folder path has, so a re-uploaded PDF is only caught per card after extraction.

Suggested change: Move scale, size limits and messages into one shared module and import it from both paths. Add the local_scan_files hash check to readUploads.

9. Approve bypasses the context gate — LOW

Where: `bulk-card-scanning.mjs:216 (PUT /api/scan-cards/:id action=approve) versus :227 and :247`

Observed: Save all and auto-save call requireContextReady, but approve checks only that an expected context exists. Cards can be approved while mismatched cards are still pending, and the batch then fails at Save all.

Suggested change: Run requireContextReady on approve as well.

10. Local scanner lock is in-process memory with no owner or timeout — LOW

Where: `local-document-scanning.mjs:13 and :16`

Observed: One variable serialises every scanner and folder job. A hung job blocks every user with "Another scanner/folder operation is running" until the process restarts, and the message does not say who started it or when.

Suggested change: Record the job id, owner and start time, show them in the message, and allow an administrator to clear a job older than a timeout.

11. Documentation and small code hygiene items — LOW

Where: `docs/vlb-elb-batch.md line 3, vlb-two-stage.mjs:8, src/bulk-card-scanning.jsx:205, log-book.mjs:118, src/vlb-entry-workspace.jsx:4`

Observed: The two-stage specification still reads "final draft, not yet built" although the feature is live by default. The operational start date 2026-11-01 is hard-coded in five UI files instead of imported from operational-start.mjs. assertVlbBatchWrite runs twice per insert. The session-storage batch memory is written and never read. The specification says reopen returns to ELB open, while a batch that was never VLB-locked returns to VLB open.

Suggested change: Update the specification status and reopen wording, import the operational start constant, drop the duplicate batch check and the unused session-storage write.

4. What was verified and how

- Code paths were traced from the Vehicle Log Book page through src/main.jsx, the entry workspace, the Manual form, the scan page, the log book API, the scan APIs, the local document scanner and the batch policy module.
- Behaviour was reasoned from source. No browser session was run against a live database.
- The full test suite was run on this checkout. All tests pass except two unrelated to these flows: the VLB batch deletion UI test fails because the Windows checkout has CRLF line endings and the test strips imports with a line-feed-only pattern, and the release packaging test fails on a temporary path. Neither indicates a product defect, but the CRLF case will fail for anyone running tests on Windows.

5. Suggested order of work

1. Picker and workspace gating on stage and amendment owner (finding 1), together with the scan batch creation check (finding 2). Both are small and remove the most wasted effort.
2. Shift lookup fix (finding 4) and inline meter warnings (finding 5) in the Manual form.
3. Team decision on auto-save (finding 3), then code or documentation change.
4. Manual-form transfer hold (finding 6) and Edit, Delete, navigation state (finding 7).
5. Shared PDF limits module and local upload dedupe (finding 8), then the low items.

6. Points for discussion

- Should a complete, clean scan card ever post without a person clicking Approve? The answer sets finding 3.
- Should a non-owner be allowed to add cards to another user's open, never-reopened batch? The code allows it today; the documentation is silent.
- Should a negative meter total block the save, or only warn and record the breach as now?